

# MagPilot | Lab update

5 October 2026 · meeting update · paper branch

### Data collection

DEMO · data\_collection/demo/ · 3 participants · 1 / 600 takes at 0 mm

Character source Sensor board Demo (mouse) Refresh

DEMO: draw with the mouse. No sensor board or robot is required; demo takes are saved separately.

### Experiment

name  Save

Included in every character recording.

### Participant

P01

name

hand

Right Left

age

not given

consent

☐

notes

excluded

☐

New Save

Demo participant 1 (P01) saved on 2026-10-05.

Digits 0-9

Letters A-J

height

0 mm 10 mm 50 mm 100 mm 150 mm 200 mm

| Participant              | Hand | A | B | C | D | E | F | G | H | I | J | Total |
|--------------------------|------|---|---|---|---|---|---|---|---|---|---|-------|
| Demo participant 1 (P01) | R    | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1     |
| Demo participant 2 (P02) | R    | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0     |
| Demo participant 3 (P03) | R    | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0     |
| All                      |      | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1     |

### Tracking error (sensor board)

magnet 12x12mm\_stack offset mm  heights mm  Start

No tracking-error runs yet.

DEMO: select 0-9 or A-J; Enter starts and saves a take. Move the mouse to draw while recording; Enter stops the ink and saves.

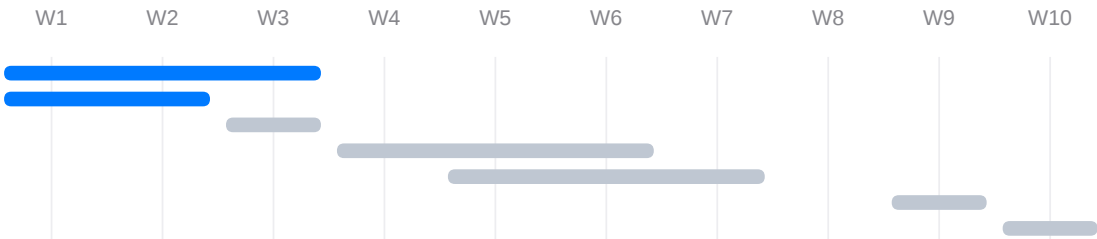
Close

DEMO · fictitious participant names; synthetic mouse input.

## Proposed ten-week plan

Relative weeks

Virtual task + pilot  
Hardware + recording  
Pilot + protocol freeze  
Full collection + preprocessing  
Models + tracking analysis  
Participant evaluation  
Analysis + report / buffer



Reference: [K&J Magnetics: magnetic moment / geometry](#)

## Collection foundations

- Protocol: 0–9 / A–J; ten takes each (200 per participant).
- Participant names and stable IDs; experiment names accompany takes.
- Enter marks capture boundaries; CSV/PNG/JSON use frozen rows.
- Separate demo files/counts; software checks + GUI rehearsal passed.

## Hardware update

- Sensor board repaired; three smaller magnets available for experiments.
- Compare one-, two- and three-magnet configurations; document geometry, grade and orientation.
- Measure signal/clipping at the writing gap and teleoperation range.

## Next milestones

- Start the virtual robot-arm target-reaching pilot now; compare completion times and endpoint errors.
- Character pilot: 40–60 takes/person (2–3 repetitions); myself + 1–2 colleagues. Not collected.
- Next: review/resume, preprocessing and participant-separated splits.

# Explicit one-character capture

Select label

&gt;

operator Enter: start

&gt;

participant draws

&gt;

operator Enter: stop/save

&gt;

inspect.

**Character collection**

**DEMO · mouse input**

Lab meeting mouse demo

Participant: P01  
Session: S01

Digits 0–9 · Letters A–J  
10 takes per character

- 1 Select the character
- 2 Position the pen / cursor
- 3 Enter to start
- 4 Move the mouse to draw
- 5 Enter to stop and save

Mouse movement draws while recording is active.

Escape cancels an unfinished take.  
Enter repeats the selected character.

**Writing Surface**

Saved letter\_A 1/10 — Enter to record another

0-9 / A-J select Enter start / stop & save Escape cancel q quit  
Move the mouse to draw while recording; Enter again repeats the selected character

DEMO · Lab meeting mouse demo | Ready: letter\_A | Enter start | Saved letter\_A 1/10 — Enter to record another

**Saved takes**

Selected: A · 1/10  
Ready · Enter starts  
Total: 1/200  
Enter start / stop · Escape cancel

0 |  
1 |  
2 |  
3 |  
4 |  
5 |  
6 |  
7 |  
8 |  
9 |  
A |  
B |  
C |  
D |  
E |  
F |  
G |  
H |  
I |  
J |

X/Y projection, 64px, trail 250, Z mode max

Actual GUI rehearsal: synthetic A; one of ten takes saved.

- Operator uses Enter for recording and demo ink; participant only moves the mouse.
- Demo mouse input; no sensor board needed.
- Stop freezes rows before matched artifacts are saved.
- Hardware pilot and new-pen validation remain pending.